IBCP YEAR 1 / ARTIFICIAL INTELLIGENCE / SUMMATIVE ASSESSMENT

Scam Shield AI

A Dialogflow ES fraud-awareness assistant with implementation and console results

STUDENT	PLATFORM	DELIVERABLE
Aarav Motwani	Google Dialogflow ES	Agent, tests and project documentation

CRS NAME	COURSE NAME
Artificial Intelligence	Artificial Intelligence: Real-World Applications and Implications

01 The problem

People who get payment requests, fake job offers, impersonation calls or unfamiliar links may react quickly without checking who is contacting them. Scam Shield AI gives advice and a safe path to official help. Scam Shield AI does not say that a message or sender is genuine.

02 Audience and design goals

The main users are people who use payments, students, job seekers, older adults and anyone who may have already shared information or sent money. The response design focuses on a warning, an explanation and a safe next step. The response does not ask users to share OTPs, UPI PINs, passwords, CVVs or full account details.

Language observations: English scenarios passed in the Dialogflow console. With Hindi selected, the recorded scam requests reached fallback or an unrelated intent.

03 Solution architecture

The agent package contains **21 intents**, including welcome and fallback. It contains **10 custom entities** and **240 example training phrases**. Each custom intent contains 12 example phrases. A required requested-action parameter shows slot filling. Short-lived contexts help with scam-detail follow-up and an OTP-safety quiz. Quick replies guide users to clear steps. The agent works without a webhook or a personal data store.

USER NEED	INTENT GROUP	SAFETY OUTCOME
Check a message	ScamCheck, ScamDetails	Pause action; describe the request without sharing secrets
Understand a scam pattern	OTP, UPI, QR, phishing, job, investment, loan, impersonation, remote access	Recognise warning signs and verify independently
Recover and report	AccountCompromised, ReportFraud, PreserveEvidence	Contact bank quickly; use reporting routes
Learn and navigate	SafePaymentTips, quiz, welcome, fallback, goodbye	Practice a safe response and recover from unclear input

DESIGN DETAIL

04 Language understanding

Custom entities identify the scam type, contact channel, requested action, credential type, payment method, claimed identity, urgency cue, suspicious-link feature, job fee and whether harm has already happened. Synonyms include variations like one-time password, UPI collect request, and registration fee. The agent stays safe even if a phrase is misclassified.

ENTITY	EXAMPLE VALUES	WHY IT MATTERS
@scam-type	UPI fraud, phishing, job scam	Routes to relevant advice
@requested-action	share OTP, send money, open link	Supports required follow-up slot
@credential-type	OTP, UPI PIN, password	Highlights secrets never to disclose
@payment-method	UPI, QR code, card	Explains payment-specific risk
@contact-channel	WhatsApp, SMS, call	Captures where contact happened
@impersonated-identity	bank, employer, support	Identifies authority claims
@urgency-cue	immediately, account will close	Recognises pressure tactics
@link-feature	shortened link, misspelled domain	Recognises suspicious links
@job-fee	registration fee, deposit	Flags upfront-payment requests
@loss-status	money sent, app installed	Prioritises urgent recovery

05 Example conversation

STEP	USER / AGENT
1. User	Can you check if this message is a scam?
2. Agent	I cannot verify that a message is safe from a description alone. Do not click links, pay or share codes. What did they ask you to do?
3. User	They want me to share my OTP.
4. Agent	That is a warning sign. Do not share the OTP. Contact the organisation using its app or website, not the sender's link or number.

06 Safety boundaries

The bot gives advice. It does not detect fraud automatically. It is not a bank representative. It never asks for secrets. It never says a transaction is safe. If money has been lost it tells the user to contact their bank. It also directs them to India's cyber-fraud helpline 1930 and cybercrime.gov.in. Advice follows NPCI and RBI fraud-awareness guidance.

DIALOGFLOW CONSOLE EVIDENCE

07 Test protocol

A live Dialogflow ES console pass covered ten cases: five scam-scenario intents, remote access, a slot/context follow-up, both quiz branches, and fallback. Three further English inputs passed on 22 October 2026.

SCENARIO	EXPECTED MATCH	OBSERVED RESULT
UPI collect request	UPICollectIntent	Matched; UPI extracted; safe warning returned
OTP impersonation	OTPIntent	Matched; OTP extracted; advised not to share it
Fake job fee	FakeJobIntent	Matched; warned about registration fees
Phishing link	PhishingLinkIntent	Matched; warned against shortened SMS link
Remote access	RemoteAccessIntent	Matched; warned against remote-control app
Money already lost	AccountCompromisedIntent	Matched; bank, 1930 and portal guidance returned
Scam detail follow-up	ScamCheckIntent -> ScamDetailsIntent	Matched; requested action captured; safe advice returned
Quiz safe answer	QuizStartIntent -> QuizSafeAnswerIntent	Matched; confirmed OTPs must not be shared
Quiz unsafe answer	QuizStartIntent -> QuizUnsafeAnswerIntent	Matched; corrected unsafe answer
Fallback	Default Fallback Intent	Matched; safe options and 1930 guidance returned

08 Recorded implementation

The Dialogflow ES console agent is ScamShieldAI (agent ID scamshieldai-k9we), with 21 intents and 10 custom entities. Ten English scenarios and three additional English inputs passed in the console. The English greeting also matched Welcome when en was selected. Recorded Hindi scam requests reached fallback or an unrelated intent. The chatbot is demonstrated through the Dialogflow ES console. The GitHub repository is private at github.com/aarav343/scam-shield-ai-sa.

09 Sources

SA brief: CRS AI Y1C1 Artificial Intelligence - Real-World Applications and Implications (Summative Assessment).

Google Cloud Dialogflow ES documentation: agent import/export and intent parameters.

Google Cloud Dialogflow Web Demo documentation: default-language limitation.

National Payments Corporation of India: fraud awareness. Reserve Bank of India: digital banking safeguards.

National Cybercrime Reporting Portal: cybercrime.gov.in and financial cyber-fraud helpline 1930.